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A Neutral and Evidence-Grounded Evaluation of Chunking Strategies in Industrial RAG Systems

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Abstract. Chunking defines the retrieval units used by retrieval augmented generation (RAG) systems, but evaluations often entangle segmentation with dataset construction. We present an evidence-grounded protocol for evaluating chunking strategies in an industrial RAG setting. The protocol generates questions from full document pages before any chunking step, anchors each question to minimal verbatim evidence snippets, and aligns those snippets to the chunks produced by each method after segmentation. This design avoids favoring the method used to create the supervision. We evaluate token-based, recursive, semantic, LLM-assisted, and adaptive chunking strategies on 165 long-form Brazilian Portuguese documents with over 3M tokens and 887 evidence-anchored questions. Retrieval-only metrics show that structure-aware and LLM-assisted chunking improve evidence coverage over semantic chunking in heterogeneous regulatory and legal corpora. Paired tests with Holm correction at $k = 10$ support the main recall differences and show no reliable difference between the LLM-assisted strategy and the adaptive decision agent. A secondary check using the RAGAS framework indicates that retrieval gains transfer to contextual precision and relevancy without reducing faithfulness.

Keywords: Retrieval augmented generation · document segmentation · chunking · information retrieval · industrial NLP

1 Introduction

Retrieval augmented generation (RAG) combines document retrieval with language generation to ground answers in external sources [13,9]. The retrieval stage usually indexes small textual units rather than full documents, so the segmentation policy that creates those units determines which evidence can be retrieved, how much surrounding context each hit carries, and how much irrelevant text enters the generator. This choice matters in industrial deployments that process contracts, regulations, audit reports, technical documents, and other long-form materials with tables, cross-references, and nested structure.

Despite this central role, chunking often remains a preprocessing default. Prior RAG work has focused on retrievers, readers, reranking, and long-context use [11,10,18,14]. Recent work has started to revisit segmentation and retrieval granularity [16,8,4,21], but a methodological problem remains: if questions or relevance labels are produced from pre-existing chunks, the evaluation can favor the segmentation that created the data. The bias is sharper in long documents, where an answer-bearing statement may cross paragraph, table, or section boundaries.

We study chunking as the sole experimental variable in a fixed RAG retrieval pipeline. The protocol first samples full document pages, generates questions from those unsegmented pages, and records minimal verbatim evidence snippets. Each chunking method then segments the same source documents. Ground truth is constructed after segmentation by matching the evidence snippets to the chunks produced by that method. All non-segmentation factors are held constant across methods.

This paper makes three contributions. First, it defines an evidence-grounded protocol for evaluating RAG chunking without deriving supervision from any candidate segmentation. Second, it applies the protocol to 165 industrial long-form Brazilian Portuguese documents and 887 questions across single-hop and multi-hop settings. Third, it compares token-based, recursive, semantic, LLM-assisted, and adaptive chunking using retrieval-only metrics, cost analysis, statistical tests, and a secondary check using the RAGAS framework.

The results show that structure-aware (recursive) and LLM-assisted strategies produce better evidence coverage than semantic chunking in heterogeneous legal and regulatory corpora. The adaptive agent approximates the LLM-assisted method while reducing the need to apply the more expensive policy to every document. These findings support treating chunking as an architectural decision in enterprise RAG systems rather than as a default preprocessing step.

2 Related Work

RAG systems retrieve external evidence and condition generation on the retrieved context [13,9]. Dense passage retrieval [11], retrieval-aware generation [10], and in-context retrieval augmentation [18] improve how systems select and use passages. Work on long-context models shows that adding more retrieved text does not guarantee better evidence use, because language models can be sensitive to the position and amount of relevant context [14]. These studies motivate retrieval units that are neither too fragmented nor too broad.

Text segmentation predates RAG and has been studied as a discourse and topic-boundary task. Bayesian and supervised methods model segmentation as the identification of coherent topical spans [6,12]. RAG changes the evaluation objective: a useful segment must be semantically coherent, retrievable for a user query, and compact enough to fit downstream context budgets. Recent work therefore studies chunking and retrieval granularity directly. Semantic chunking evaluations question whether embedding-based boundaries justify their computational cost [16]. Late chunking delays pooling until after long-context

encoding to preserve surrounding context [8]. Proposition-level and atomic-unit retrieval index finer factual units instead of passages [4,17]. Hierarchical approaches such as RAPTOR and HiChunk build retrieval structures above flat chunks [19,15]. PIC and MoC further show that document segmentation affects both retrieval and downstream QA metrics [21,23].

Closer to our setting, Brazilian Portuguese work on legal and regulatory documents motivates segmentation policies that follow functional units. Rhetorical-role identification for Portuguese legal petitions shows that legal documents organize content into functional units such as facts, legal bases, precedents, requests, and remedies [1]. TableRAG adapts RAG to retrieved tabular evidence, indicating that layout structures are part of the evidence surface in formal documents [20]. The closest prior approach is a two-stage architecture for NDA analysis that uses LLM-based segmentation before clause classification [3]; we differ in evaluating segmentation as the experimental variable in retrieval rather than using it as a preprocessing step for downstream classification. None of these works compares general chunking policies under a fixed retriever, but they support the premise that useful retrieval units in formal documents often align with functional or layout-based boundaries.

Our work is complementary to the chunking and retrieval-granularity work above. We do not propose a new retriever or a new retrieval architecture. Instead, we evaluate chunking policies under a fixed embedding and retrieval configuration so that observed differences can be attributed to segmentation. The closest methodological concern appears in evidence-grounded QA and multi-hop retrieval, where answer quality depends on retrieving all relevant support [22,2,5]. We adopt that evidence focus but avoid generating supervision from chunks. This separation lets us compare chunking strategies without introducing segmentation-induced bias.

3 Evaluation Protocol

3.1 Problem Formulation

Let $D = \{d_1, \ldots, d_n\}$ denote a corpus of long-form documents. A chunking strategy C maps each document d_i to a sequence of chunks $C(d_i) = \{c_{i1}, \ldots, c_{im}\}$. Let $\mathcal{I}_C$ denote the index containing all chunks produced by C. A fixed embedding model E maps each query q and chunk c to dense vectors, and sim denotes cosine similarity. Retrieval ranks chunks by similarity:

$$R_k(q, C, D) = \text{top}_k \left\{ \text{sim}(E(q), E(c)) : c \in \mathcal{I}_C \right\}. \tag{1}$$

The experiment varies C while keeping the embedding model, the index implementation, the similarity function, and the retrieval cutoff fixed. This isolates segmentation from other RAG design choices.

3.2 Corpus and Question Construction

The corpus contains 165 Brazilian Portuguese PDF documents converted to Markdown with an LLM-assisted extraction pipeline (Gemini 2.5 Flash). We

Table 1. Document distribution and predominant structure in the evaluation corpus.

Document type	Count	Predominant structure
Legislation	57	Hierarchical clauses and delimiters
Legal documents (electricity sector)	29	Narrative reasoning and cross-references
ANEEL transmission reports	14	Quarterly reports with prose, tables, and annexes
General reports	33	Sectioned institutional reports
Theses and dissertations	32	Long academic exposition
Total	165	3,013,692 tokens

cap documents at 40 pages to bound extraction cost while keeping multi-section structure within reach. Table 1 summarizes the corpus. The mix of legislation, legal reasoning, ANEEL transmission reports[3], general institutional reports, and theses creates variation in hierarchy, cross-references, tables, and narrative density.

Question generation occurs before chunking. The pipeline first selects information-bearing excerpts from full pages with Gemini 2.5 Flash Lite, then drafts a self-contained question and concise answer with Gemini 2.5 Flash, and finally validates each item with Gemini 2.5 Flash Lite, allowing up to two refinement iterations when validation fails. For every accepted item, the pipeline stores minimal verbatim snippets that justify the answer. The resulting set contains 887 questions, divided into single-hop questions whose evidence appears in one local passage and multi-hop questions whose evidence requires combining snippets from different passages. Because evidence snippets are extracted from full pages rather than chunks, no segmentation method contributes to the supervision.

3.3 Chunking Strategies

Table 2 lists the strategies. All chunkers are instantiated through the Chonkie library[4] with shared retrieval components. Token-based chunking uses fixed-size windows and therefore controls index size but ignores document structure. Recursive chunking follows Markdown hierarchy before falling back to paragraphs and sentences. Semantic chunking uses E5-small sentence embeddings to detect local topic shifts. The LLM-assisted strategy, `slumber`[5], first creates short recursive candidate segments and then asks an LLM to merge adjacent candidates when a wider unit better preserves evidence. The adaptive decision agent is a production-oriented routing policy: for each document, it chooses either recursive chunking, which is fast and has no external LLM cost, or `slumber`, which is slower but can preserve evidence in less regular documents.

[3] ANEEL is the Brazilian regulator for the electricity sector; the reports are quarterly transmission filings (RAD).

[4] `https://docs.chonkie.ai/oss/chunkers/overview`

[5] `slumber` is Chonkie's agentic chunker.

Table 2. Chunking configurations. Shared retrieval uses `gemini-embeddings-01`; token-based methods use the `o200k_base` tokenizer.

Strategy	Size	Configuration
Token-based	512	Fixed window; overlap 128; `o200k_base`
Recursive	512	Markdown; `o200k_base`
Semantic	512 max	E5-small; threshold 0.5; window 3
`slumber`	1024	Recursive candidates; candidate size 128
Decision agent	512 or 1024	$\lceil 1.5\sqrt{n} \rceil$ samples; recursive or `slumber`

The semantic chunker uses `intfloat/multilingual-e5-small`, a similarity window of three sentences, a minimum of three sentences per chunk, and adjacent sentence comparisons only. The LLM-assisted merge prompt receives an ordered window of 128-token candidate segments and asks for JSON output with the adjacent groups that should remain together. The instruction prioritizes preserving legal clauses, table captions with their rows, enumerated lists, and cross-referenced passages, while rejecting merges that combine unrelated topics. We use Gemini 3 Flash Preview with temperature 0.2. The same prompt template is used for all documents.

The decision agent reduces LLM use by sampling $\lceil 1.5\sqrt{n} \rceil$ representative content units from a document with n available units, such as pages or extracted page-level segments; the multiplier was set empirically in pilot runs as the smallest value that stabilized the agent's choice. The prompt presents only two available actions, recursive and `slumber`, because they represent the main quality-cost alternatives in our pipeline: recursive is a no-cost structural baseline, while `slumber` pays for content-aware boundary decisions. The prompt asks the model to inspect whether visible structure is reliable, whether long sections contain topic shifts, whether tables and enumerations need to remain attached to surrounding prose, and whether formatting noise may break rule-based splitting. It then requires a JSON response with a short rationale and `chosen_method` set to either `recursive` or `slumber`. We use Gemini 3 Flash Preview with temperature 0.2 and extended reasoning enabled. This policy makes the agent a cost-aware selector rather than a separate segmentation algorithm.

3.4 Alignment and Metrics

For each question and chunking strategy, the evidence snippets are aligned to the produced chunks after segmentation. Exact string matching is attempted first. If extraction artifacts prevent an exact match, fuzzy matching using `difflib.SequenceMatcher` with a similarity ratio of 0.8 is used as a fallback. All chunks that contain a supporting snippet are marked as relevant. This matters for multi-hop questions: if evidence is split across chunks, every evidence-bearing chunk becomes part of the ground truth set.

We evaluate Hit@k, Recall@k, MRR@k, and MAP@k for $k \in \{1, 2, 3, 5, 10\}$. Hit@k measures whether any relevant chunk is retrieved, Recall@k measures evidence coverage, and MRR@k and MAP@k account for ranking. For omnibus

tests across methods, we use Cochran's Q for the paired binary Hit@k outcomes and Friedman tests for the continuous metrics. For pairwise comparisons we use McNemar tests for Hit@k and paired bootstrap tests with 10,000 resamples for Recall@k, MRR@k, and MAP@k. Pairwise p-values are corrected with the Holm procedure.

4 Results

4.1 Retrieval Effectiveness

The retrieval-only evaluation at $k = 5$ serves as the primary operating point because it reflects a common production budget for retrieved context. Figure 1 reports Hit@5, Recall@5, MRR@5, and MAP@5 by method and question type. The LLM-assisted `slumber` strategy obtains the best aggregate Hit@5 and Recall@5, with 0.820 Hit@5 and 0.775 Recall@5. The decision agent and recursive chunking form the next tier, while semantic chunking has lower Hit@5 and Recall@5, 0.732 and 0.652, respectively. This pattern indicates that semantic sentence similarity alone does not reliably preserve evidence units in documents with clauses, tables, and cross-references.

The same ordering appears at the stricter $k = 1$, $k = 2$, and $k = 3$ diagnostic cutoffs, where small boundary errors have stronger effects on Hit@k and MRR@k. Moving from these stricter cutoffs to $k = 5$ mainly increases coverage; it does not reverse the ranking of the main strategies.

The single-hop and multi-hop split clarifies the source of retrieval failures. Single-hop questions usually require preserving one answer-bearing span. Multi-hop questions require all supporting snippets to appear in the retrieved set, so they penalize boundary decisions that separate related clauses or distribute table context across chunks. At $k = 5$, Recall decreases by about 0.11 to 0.13 for multi-hop questions relative to single-hop questions, depending on the strategy. The gap is measurable but not a collapse, which suggests that many failures come from ranking and partial evidence coverage rather than complete absence of relevant chunks.

Domain-level analysis shows the same ordering with different absolute difficulty. Structured theses and institutional reports approach ceiling retrieval for several strategies. Legal and regulatory documents, especially the ANEEL transmission reports, produce lower scores because answer evidence often spans clauses, annexes, and references. This supports the interpretation that chunking matters most when the document layout does not align cleanly with semantic evidence.

4.2 Statistical Tests

Figure 2 shows that the same ordering largely persists at $k = 10$: increasing the retrieval budget raises all methods, but it does not remove the gap between semantic chunking and structure-aware or LLM-assisted strategies. Table 3

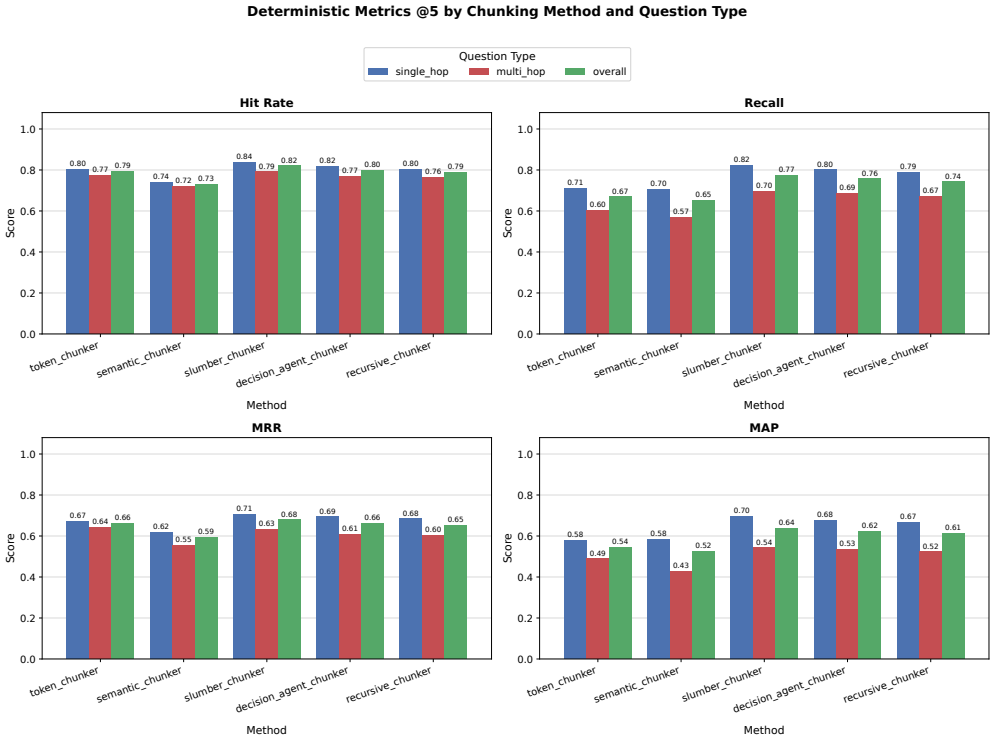


Fig. 1. Retrieval-only metrics at $k = 5$ across chunking strategies and question types. The operating point exposes the trade-off between early evidence coverage and ranking quality under a bounded retrieval budget.

summarizes the paired statistical analysis at $k = 10$. Friedman and Cochran's Q tests reject equality across methods for all four metrics. Pairwise tests show that `slumber` and the decision agent outperform semantic chunking on Hit@10 and Recall@10 after Holm correction. `slumber` and the decision agent do not differ reliably from each other on Hit@10, Recall@10, MRR@10, or MAP@10. This result supports the cost-aware routing claim: the agent preserves most of the LLM-assisted retrieval benefit while avoiding universal LLM-assisted chunking.

4.3 RAGAS Check and Cost

Because the main protocol intentionally isolates retrieval, we also ran a secondary check using the RAGAS framework [7] to see whether retrieval differences propagate to generation-oriented indicators. Figure 3 shows that methods with stronger retrieval-only metrics also improve contextual precision and contextual relevancy, while faithfulness remains high. At the aggregate level, contextual precision increases from 0.648 with semantic chunking to 0.754 with `slumber` and 0.736 with the decision agent; faithfulness remains close across these methods,

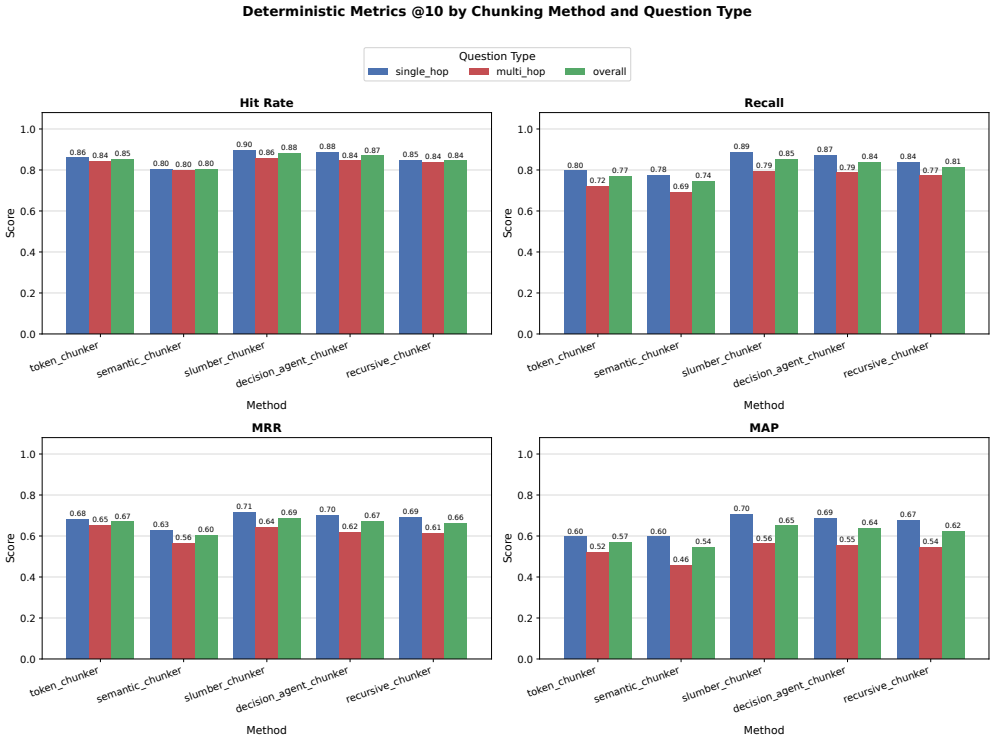


Fig. 2. Retrieval-only metrics at $k = 10$ across chunking strategies and question types. The larger cutoff improves absolute coverage but preserves the main method ordering.

between 0.927 and 0.949. We treat this result as a consistency check rather than as the main claim, because RAGAS introduces its own LLM-based judgments and does not replace controlled retrieval metrics.

Table 4 reports the offline chunking and index-preparation profile for all methods. These values do not include answer-generation cost at query time. Applying `slumber` to all documents costs US$2.21 and takes 431.21 minutes, while the decision agent reduces this to US$1.90 and 271.01 minutes by routing part of the corpus to recursive chunking. The decision agent also produces more chunks than `slumber` alone, but keeps larger average chunks than semantic chunking. The relevant trade-off is therefore retrieval quality versus offline preparation cost, preparation latency, and index granularity. Costs reflect the public Gemini API pricing at the time of the experiments.

5 Discussion and Conclusion

The evaluation shows that chunking is a first-order design choice with measurable effects on industrial RAG retrieval. Fixed semantic boundaries underperform

Table 3. Selected pairwise results at $k = 10$ with Holm-corrected p-values. Positive differences favor method A.

Metric	Method A	Method B	Difference	p_{Holm}
Hit@10	`slumber`	semantic	0.1146	0.000046
Hit@10	decision	semantic	0.1146	0.000606
Recall@10	`slumber`	semantic	0.1471	< 0.0001
Recall@10	decision	semantic	0.1375	< 0.0001
Recall@10	`slumber`	decision	0.0096	0.6108
MAP@10	`slumber`	semantic	0.0788	0.0276

Table 4. Offline chunking and index-preparation profile for 165 documents. Costs are reported only for LLM calls used by `slumber` and the decision agent. Time for the semantic chunker includes local inference of multilingual E5 sentence embeddings.

Strategy	Time (min)	Chunks	Tokens/chunk	LLM calls	Cost (US$)
Token	2.02	7241	508±32	0	0.000
Recursive	7.14	7790	356±147	0	0.000
Semantic	66.23	14413	192±177	0	0.000
`slumber`	431.21	5644	492±282	5644	2.214
Decision agent	271.01	6341	437±237	3533	1.904

when documents contain legal clauses, tables, enumerations, and cross-references. Recursive chunking is a strong baseline when visible hierarchy is reliable. LLM-assisted chunking improves evidence preservation in heterogeneous layouts, and the decision agent offers a practical compromise by applying that costlier strategy only when structural cues are weak.

The protocol also addresses a source of evaluation bias. By generating questions from full pages and aligning verbatim evidence snippets after segmentation, the benchmark does not assume that any candidate chunking strategy defines the correct ground truth. This design makes retrieval metrics easier to interpret: a method performs better when it retrieves the same independently defined evidence, not when the dataset was built in its preferred units.

The study has limitations. The retrieval layer uses a single embedding family (`gemini-embeddings-01`) and a single index implementation, so the findings may change with late-interaction retrievers, hybrid sparse-dense search, or domain-tuned embeddings. The corpus is in Brazilian Portuguese and reflects industrial legal, regulatory, technical, and academic documents, but it is not a public benchmark; results may differ in monolingual English or other-language settings. The question generation pipeline also uses LLMs, which may introduce phrasing biases even though the evidence snippets and alignment are reference-based. Finally, the RAGAS experiment is a secondary validation rather than a full human evaluation of answer quality.

Three threats to validity deserve explicit emphasis. First, the protocol is neutral with respect to the candidate chunking strategies, because the supervision is fixed before segmentation, but it is not independent of generative-model behav-

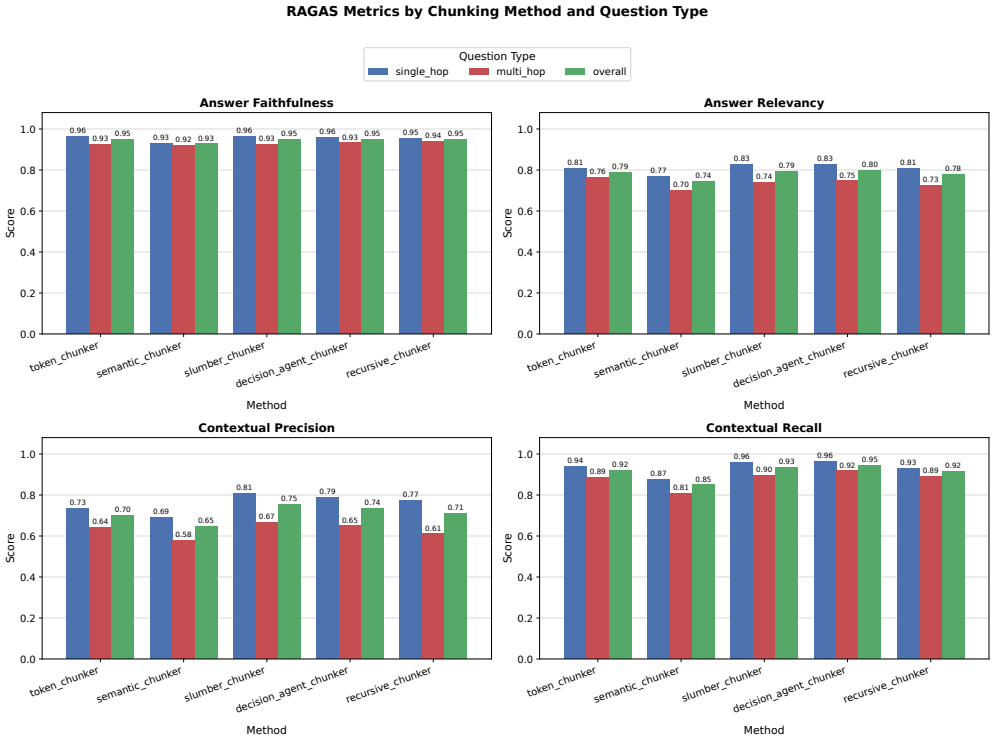


Fig. 3. Secondary RAGAS comparison across chunking strategies. Retrieval gains align with higher contextual precision and relevancy, while faithfulness remains stable.

ior. The same model family supports document extraction, question generation, the LLM-assisted chunker, and the decision agent, so part of the advantage of the LLM-based strategies could reflect representational regularities shared with the question-generation and embedding models. Validating the protocol across independent model families is a natural next step. Second, chunk size is itself a property of each strategy, and the evaluation at a fixed top-k reflects the realistic deployment condition. Because the strategies still retrieve different amounts of text (semantic near 192 tokens against 492 for `slumber`), an evaluation at matched token budgets would separate boundary quality from context volume more cleanly. The gains in MRR and MAP, together with the higher contextual precision reported by RAGAS, which a volume-only effect would not produce, indicate that the advantage does not follow from retrieved volume alone. Third, the paired tests treat questions as independent units, although questions are nested within documents, so observations from the same document are correlated. A clustered analysis would refine the reported confidence levels, even though the main differences hold across retrieval cutoffs.

Future work should test the protocol with additional retrievers, public corpora, and end-to-end answer evaluations. It should also compare flat chunking with hierarchical and proposition-level retrieval architectures under the same evidence-grounded supervision. Within the present constraints, the results support a pragmatic recommendation: production RAG systems should document and evaluate chunking choices with the same care given to embedding models and rerankers.

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